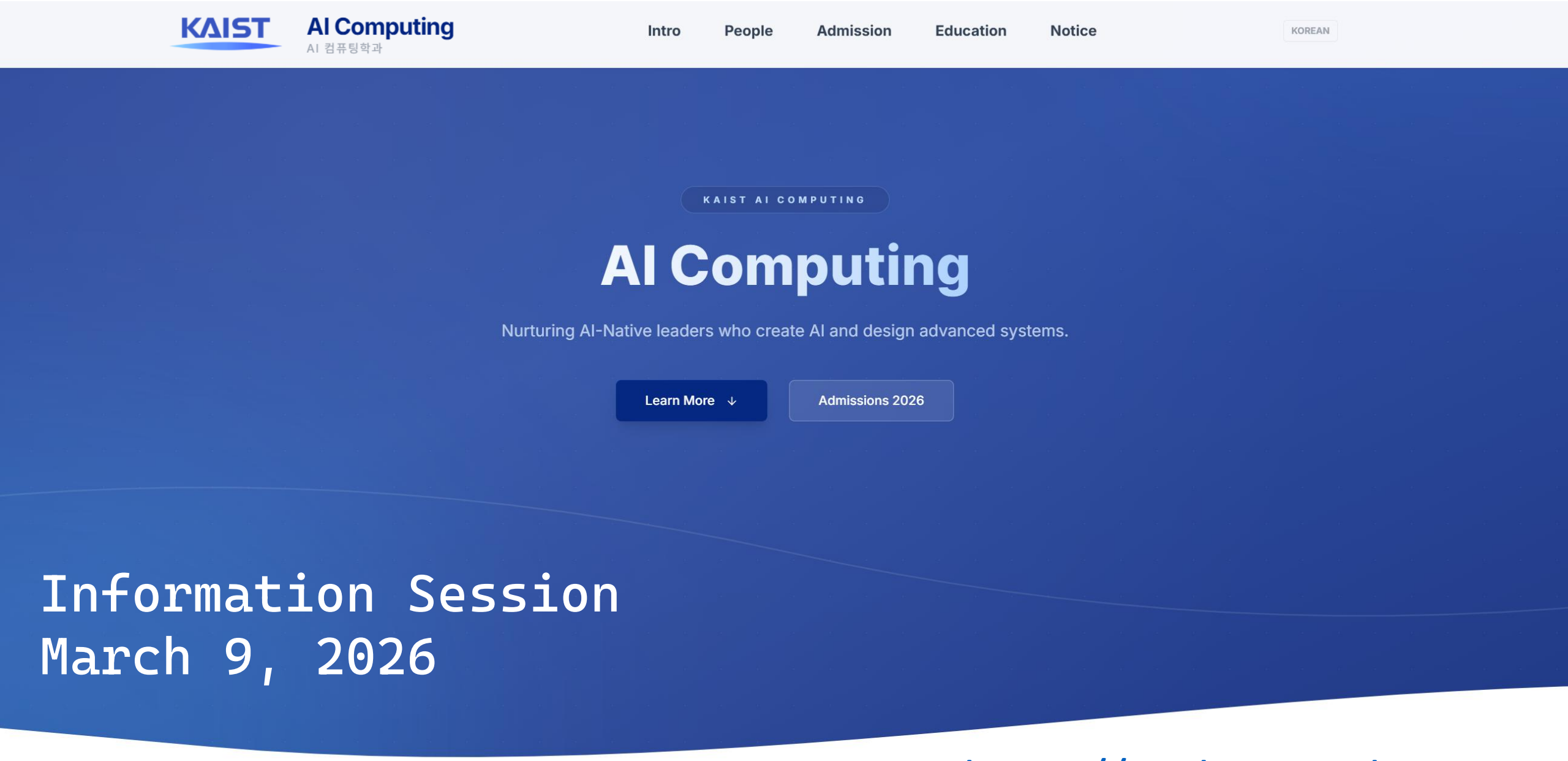


Information Session
March 9, 2026

https://aic.kaist.ac.kr

Computing Trend

Make Tech Functional 기능성 → **Make Tech Usable 사용성** → **Make Tech Persuasive 설득성** → **Make Tech Symbiotic 공생성**

1970s — 1990s — 2010s — 2020s–

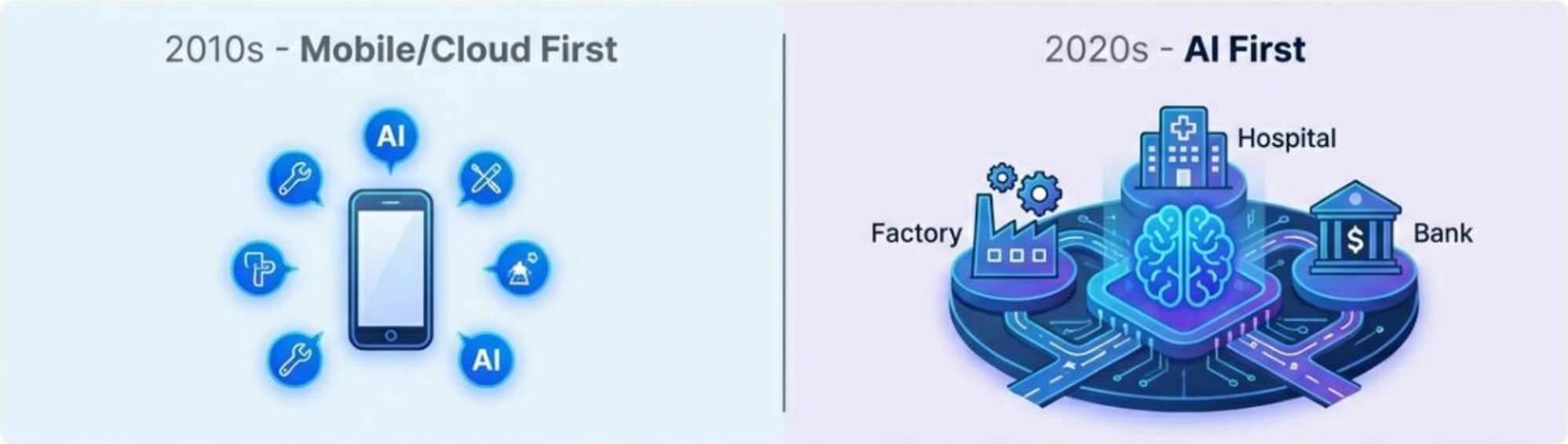
Main Frame

Personal Computers

Mobile, Wearable, IoT Devices

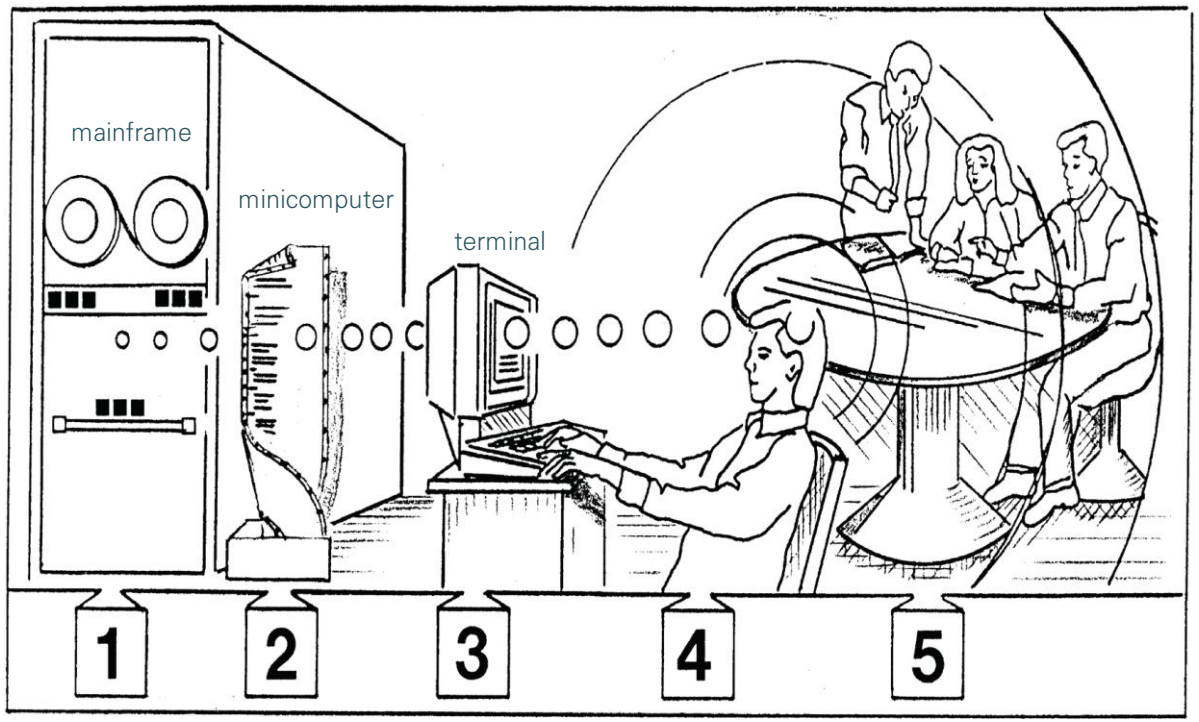
AI Agents, Physical AI

Industry Shifts: Mobile to AI



AI-Driven Industry	New Competitive Paradigm	National AI Assets
AI-centered redesign of industrial value chains	From AI adoption → AI internalization & design	Infrastructure · Algorithms · Data

Computing Paradigm Shift: From Tools to Partners



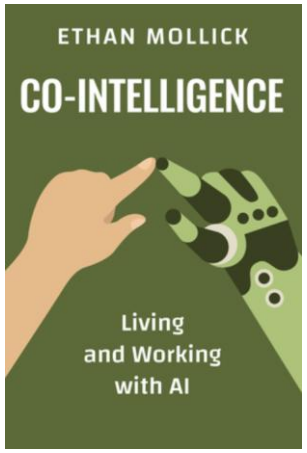
Interface as Hardware 인터페이스: 하드웨어	Software 소프트웨어	Terminal 터미널	Dialogue 대화	Social & Ubiquitous 소셜 및 유비쿼터스

+

perplexity
ChatGPT
Gemini
Anthropic Claude AI
Copilot

From Tool to Partner: The Evolution of Human -Computer Interaction, Jonathan Grudin, 2018

Erik Brynjolfsson
Andrew McAfee

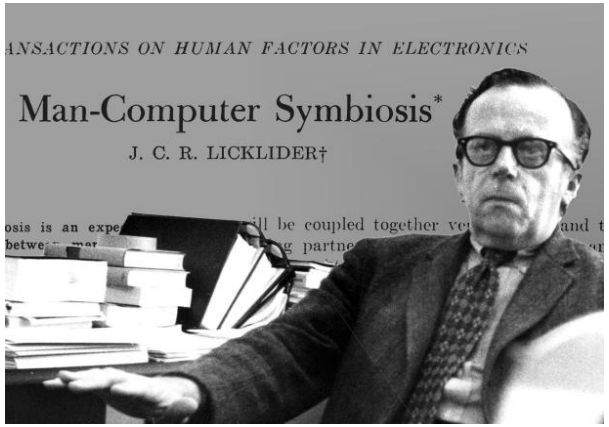


Ethan
Mollick

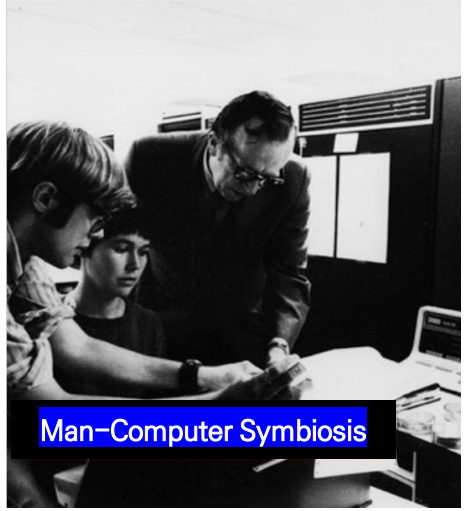
Automation in the Machine Age: In the machine age, many routine and repetitive tasks are automated (e.g., autonomous vehicles, question answering, personal assistants).

Beyond Routine Tasks: Humans are capable of creative idea generation, broad reasoning, and complex communication.

Co-Intelligence (Dual Brain): Through human-machine collaboration, limitations can be overcome; e.g., Human physician + AI agent



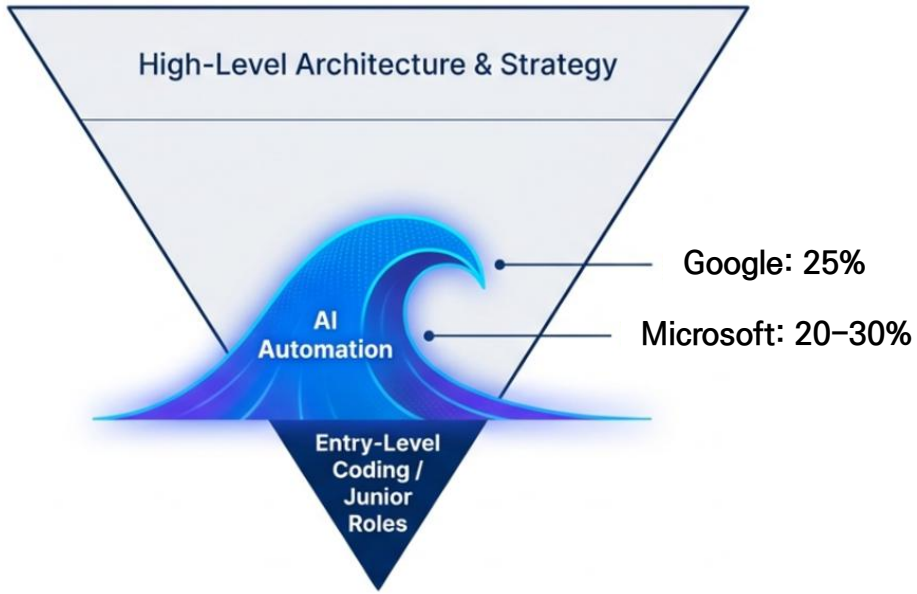
Man-Computer Symbiosis
J.C.R. Licklider 1960



"In the anticipated partnership between humans and computers, humans will set goals, formulate hypotheses, determine criteria, and perform evaluations. Computing machines will carry out the routine operations necessary for technical and scientific thinking. Preliminary analysis suggests that such a cooperative partnership will perform intellectual work much more effectively than humans alone."

AI is not ending programming — it is reinventing it

- Every major abstraction shift in history created more programmers, not fewer
- AI enables natural-language coding, but **expertise is still required** for design, debugging, and edge cases
- The **role shifts** from **writing code** to **systematic thinking and managing AI systems**
- The future belongs to developers who collaborate with AI, expanding what is programmable



The End of Programming as We Know It, By Tim O'Reilly February 4, 2025
https://www.oreilly.com/radar/the-end-of-programming-as-we-know-it

New Education Demand

Expanding AI Education

Growing industry demand and investment are driving universities to expand AI education

Rapid Growth of AI Majors

At leading universities, such as MIT, AI has become one of the fastest-growing majors

AI Undergraduate Programs in Korea

New AI undergraduate programs (e.g., KAIST, Yonsei, Korea University)

The New York Times

Grok Controversy | Lawsuit Over A.I. Hiring Decisions | Electricity Costs | Data Centers in Space? | Kille

College Students Flock to a New Major: A.I.

At M.I.T., a new program called "artificial intelligence and decision-making" is now the second-most-popular undergraduate major.

By Natasha Singer

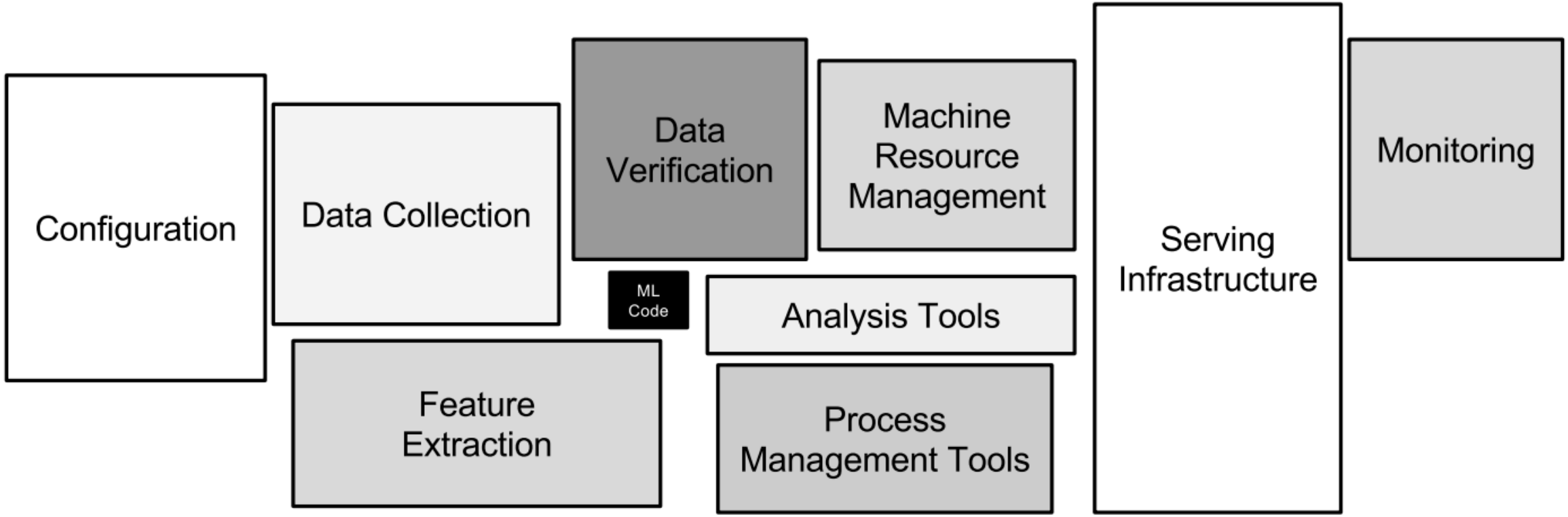
Natasha Singer covers computer science and A.I. education.

Dec. 1, 2025

AI Computing – Full Stack AI Expert

Hidden Technical Debt in Machine Learning Systems, Sculley et al., NeurIPS 2015

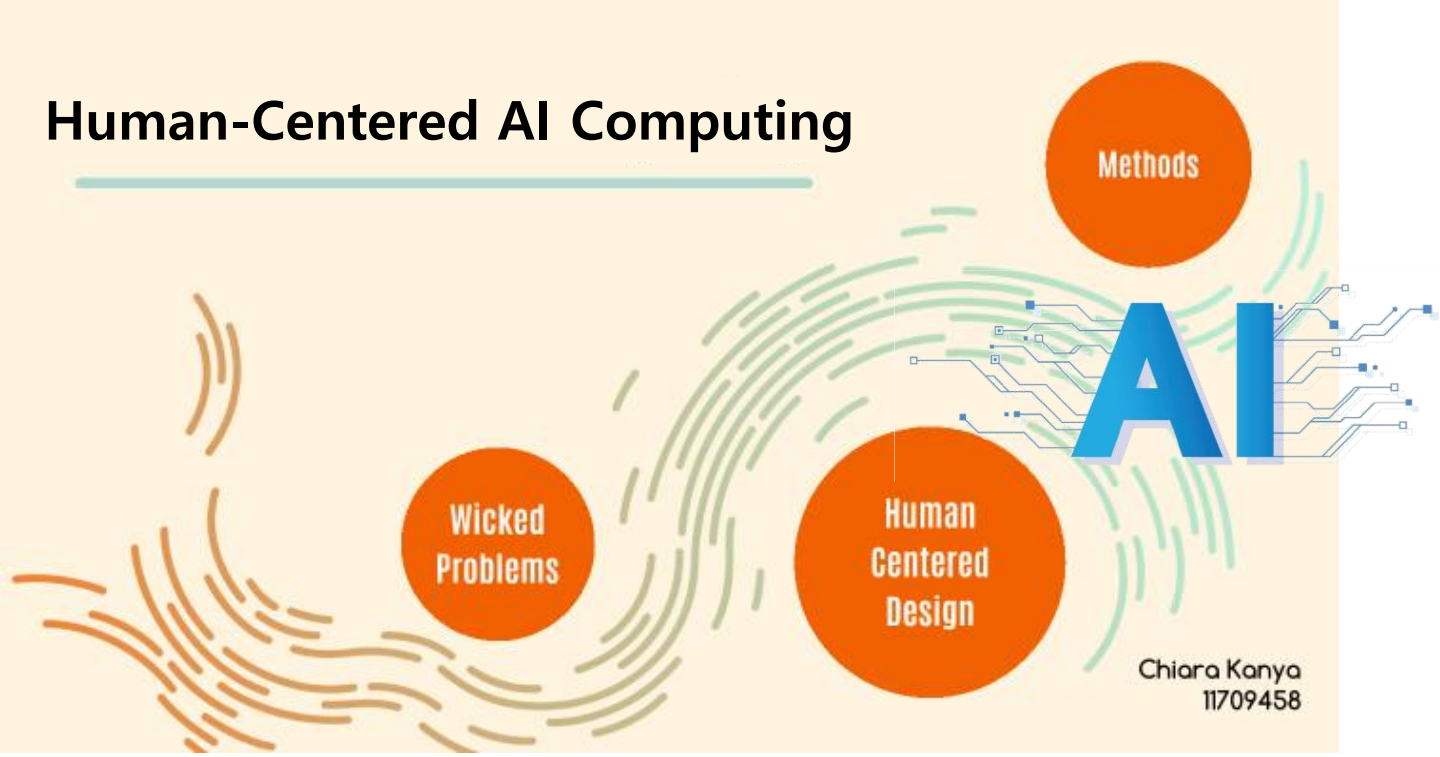
AI Computing – Full Stack AI Expert



Hidden Technical Debt in Machine Learning Systems, Sculley et al., NeurIPS 2015

KAIST AI Computing Department Vision
Human-Centered AI Computing

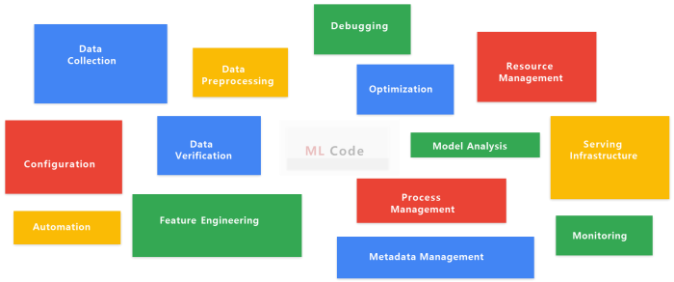
An interdisciplinary field focused on creating AI computing systems that support and enhance human activities.



AI IN CS

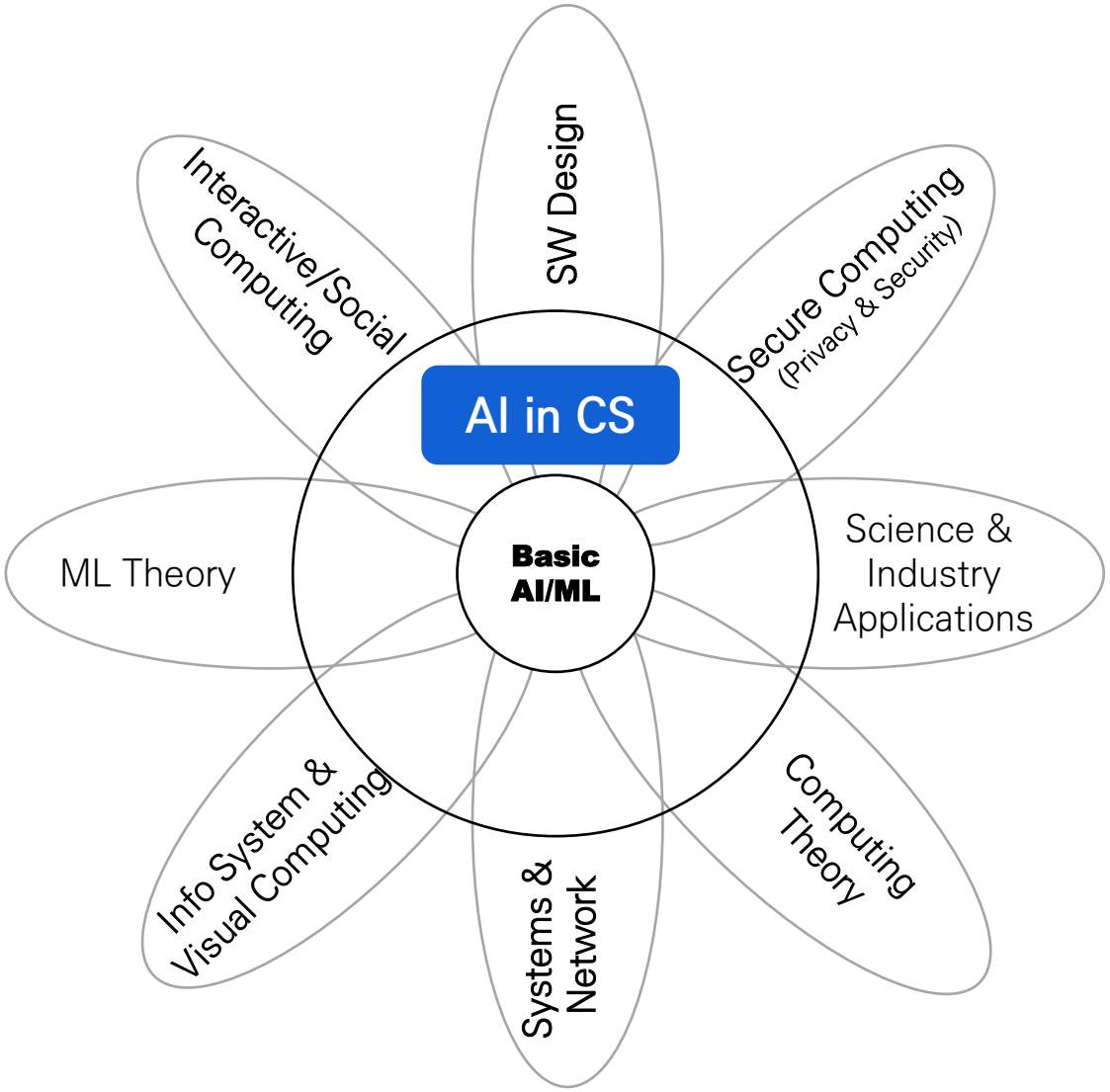
https://lastfuturist.com/what-is-human-centric-ai/

Human-Centric AI Computing



Beyond ML Code –
AI Systems & Applications
"Hidden Technical Debt in Machine Learning Systems, Sculley et al., NeurIPS 2015"

* Kim Jaechul Grad School of AI



* AX Dept.
* Grad School of Data Science

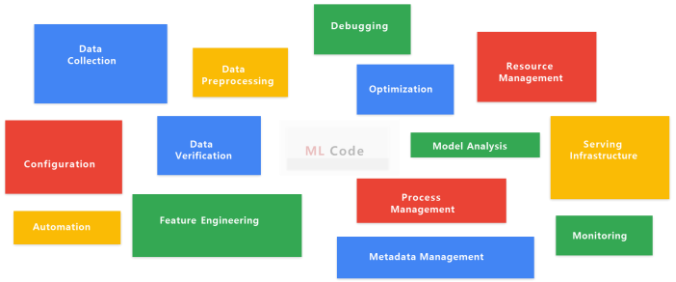
* AI Systems Dept.

AI COMPUTING

SPECIALIZED PROGRAM FOR HUMAN-CENTERED AI COMPUTING

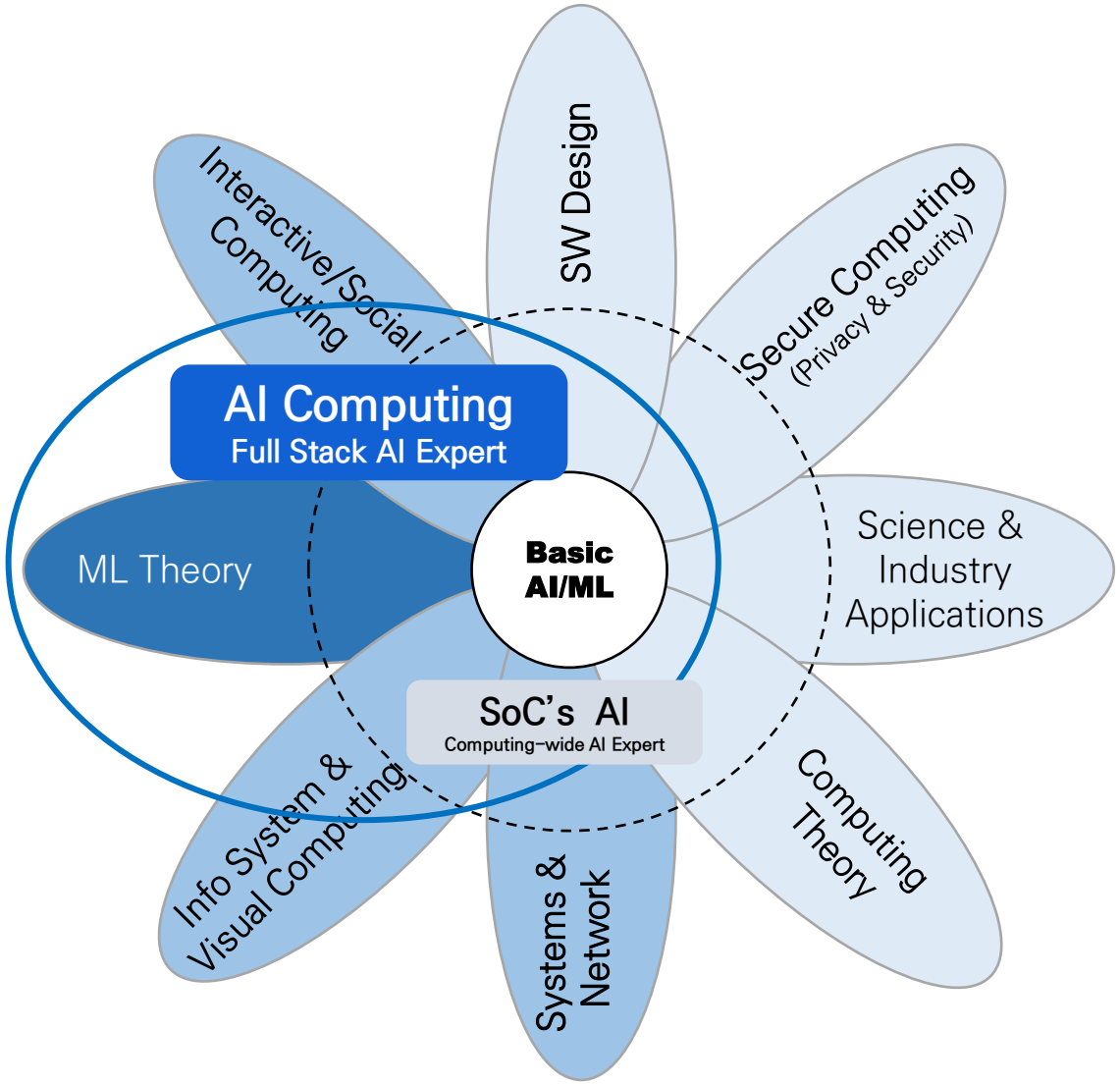
https://lastfuturist.com/what-is-human-centric-ai/

Human-Centric AI Computing



Beyond ML Code –
AI Systems & Applications
"Hidden Technical Debt in Machine Learning Systems, Sculley et al., NeurIPS 2015"

* Kim Jaechul Grad School of AI



* AX Dept.
* Grad School of Data Science

* AI Systems Dept.

https://aic.kaist.ac.kr

Global Leading Faculty of 29 Members

Uichin Lee
CORE FACULTY, CHAIR

Minhyuk Sung
CORE FACULTY

Sungjin Ahn
CORE FACULTY

Seunghoon Hong
CORE FACULTY

Joyce Jiyoung Whang
CORE FACULTY

Tae-Kyun Kim

Hyunwoo J. Kim

Sukyoung Ryu

Noseong Park

Daehyung Park

In-Young Ko

Youngjin Kwon

Daeyoung Kim

Min-Soo Kim

Juho Kim

Jongse Park

Jinah Park

Jongmoon Baik

Sooel Son

Joseph J. Seering

Tae-Kyun Kim

Hyunwoo J. Kim

Sukyoung Ryu

Noseong Park

Daehyung Park

Junehwa Song

Tae-Hyun Oh

Alice Oh

Shin Yoo

Sung-Eui Yoon

Jongse Park

Jinah Park

Jongmoon Baik

Sooel Son

Joseph J. Seering

Geehyuk Lee

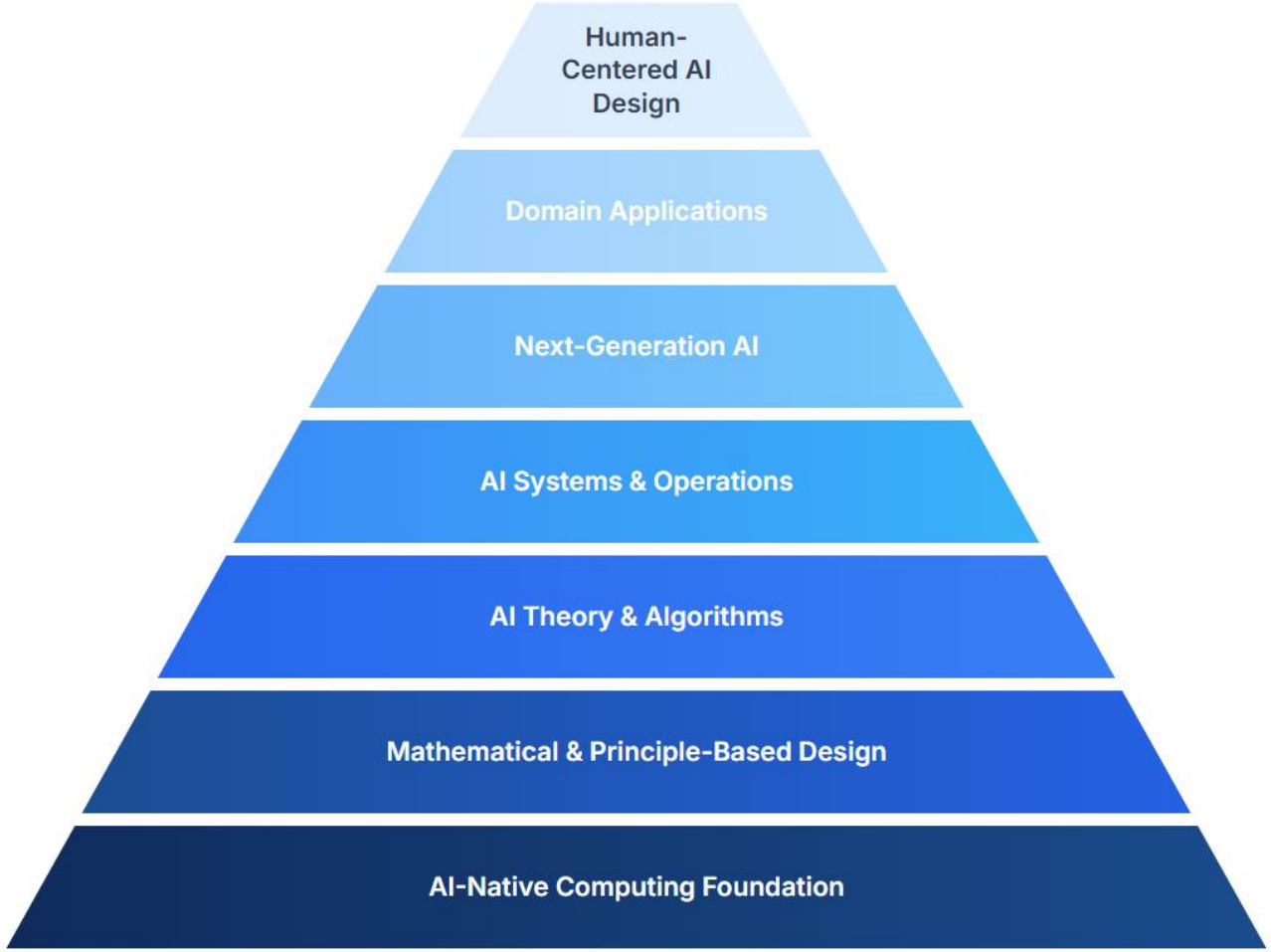
Jae-Gil Lee

Sungho Jo

Jun Han

AI-Native Education for Future Leaders

Educating AI-Native leaders for the era of Human-Centered AI Computing



AI-NATIVE CORE COMPETENCIES

Responsible AI
AI Design Grounded in Ethics and Philosophical Principles

Domain Application
AI+X Solution Across Industry, Science, and Public Sectors

Next-Generation AI
Leading Innovation in LLM, Generative, and Agentic AI

AI System Engineering
GPU programming, MLOps & distributed AI systems optimization

AI Design Capability
Deep understanding of AI principles and designing new model

Undergraduate Degree

- Major, Double Major, Minor Requirements
  - Major (전공) – 44 credits
    - 12 credits of required major courses + 32 credits of major electives
  - Double Major (복수전공) – 40 credits
    - 12 credits of required major courses + 28 credits of major electives
  - Minor (부전공) – 21 credits
    - 12 credits of required major courses + 9 credits of major electives
- Curriculum – Major Courses
  - AI Mathematics: Mathematics for AI
  - AI Foundations: Introduction to AI, Machine Learning, Deep Learning
  - AI Algorithms: Reinforcement Learning, Graph Neural Networks, Natural Language Processing, Computer Vision
  - AI Systems: GPU Programming, Systems for AI
  - AI + Human: Human–AI Interaction, Wearable Interfaces
  - AI Philosophy & Ethics: Responsible & Trustworthy AI
  - AI Capstone Project

Department of AI Computing | Undergraduate Curriculum Roadmap

AIC100 Elements of AI → AIC200 Introduction to AI (Required)

AIC100 AI Philosophy (Required)

AIC200 Mathematics for AI → AIC300 Statistical Learning Theory

AIC200 Machine Learning (Required) → AIC300 Introduction to Bayesian Machine Learning

AIC300 Optimization Methods for AI

AIC300 Introduction to Reinforcement Learning

AIC400 Graph Machine Learning and Mining

AIC400 Brain-Inspired Machine Intelligence

Tracks

- AI Foundation
- AI Algorithm
- AI + Human
- AI System

Required Courses

- AI Philosophy (AIC100)
- Introduction to AI (AIC200)
- Introduction to Deep Learning (AIC200)
- Machine Learning (AIC200)

AI Foundation

AIC200 Introduction to Deep Learning (Required)

AI Algorithm

AIC300 Introduction to Large Language Models → AIC400 AI Agent System

AIC300 Introduction to Computer Vision → AIC400 Machine Learning for 3D Data

AIC300 Introduction to Generative Models

AI + Human

AIC200 Human-Computer Interaction → AIC300 Human-AI Interaction → AIC400 Wearable User Interface

AIC400 Introduction to Social Computing

AIC400 Responsible AI

AI System

AIC200 System Programming → AIC300 Operating Systems and Lab → AIC400 System for AI

AIC300 GPU Programming

AIC300 Computer Architecture

Capstone

AIC400 AI Capstone Project

KAIST College of AI AI대학

** This course tree is tentative and please see the up-to-date info at the home page*
** Until Fall 2026, please take substitute courses in CS*
** As of Fall 2026, students should take AIC code courses*
** See the details: aic.kaist.ac.kr*

Graduate Program in AI Computing Department

- AI Computing: Admitting approximately 30 students (Fall 2026)
- Proposed Graduate Curriculum
  - **Advanced AI Theory & Algorithms**: AI & ML, Reinforcement Learning, Advanced ML
  - **Advanced AI Systems**: Distributed AI, Large-Scale Model Training, System Architecture Design
  - **Resource-Efficient AI Technologies**: Edge AI, Model Compression
  - **AI + Human (Advanced)**: Advanced Human-AI Interaction
  - **AI Ethics & Safety**: AI Model Safety Evaluation, Privacy and Security Issues
  - **Emerging AI Topics**: Large Language Models, Vision-Language Models, Generative Models, Multimodal Learning, Retrieval-Augmented Generation (RAG)
  - **Emerging Research Areas**: Agentic AI, Physical AI
  - **AI Research Seminars**: Latest research presented at top-tier conferences in AI, ML, NLP, and Computer Vision
  - **Industry Internship (4-6 months)**: Hands-on experience applying research outcomes to real-world problems in leading AI companies

Q&A

- CS ⇔ AIC (Major/Double-Major) – Double Counting Not Permitted
  - Currently no courses eligible due to requirement complexity (전산 ⇔ AIC – 복수전공 중복인정 없음)
  - Will consider this in the future when AI Computing courses are stabilized (no guarantee though)
- Differences between AI Specialized Minor (AI 특별지정전공) vs. AI Computing Minor
  - AI Specialized Minor: 18 credits = 6 required + 12 electives
    - Required Major (전공필수): Programming Basics + AI Basics
    - Designed Electives (지정선택): NLP, CV, Robotics, Advanced ML, Data Science, AI in Society
    - Electives (선택과목): X+AI, AI Basics
  - AI Computing Minor: 21 credits = 12 required + 9 electives
    - Required Major (전공필수): AI Philosophy and Ethics, Intro to AI, ML, Deep Learning
    - Major Electives (전공선택): Any major courses in AI Computing
    - More specialized for Core AI Computing subjects
- Grad School Minor (AI대학 대학원 부전공 프로그램): No plans yet

KAIST AI Computing
AI 컴퓨팅학과

Intro People Admission Education Notice KOREAN

AI-Native Education for Future Leaders

Educating AI-Native leaders for the era of Human-Centered AI Computing

https://aic.kaist.ac.kr